Original run: results at a glance

Scan 1171324 | UID 7cd58d5e-7f31-4fe4-b64f-aad0bde951db

Run completed with exit status 'success' in 6.55 hours. The acquisition produced useful high-energy thickness estimates and beam-profile diagnostics, but it did not establish a uniformly validated calibration across all three energies.

Energy	Completed / planned	Skipped	Fit rank	Reduced chi-square
8.8 keV	29 / 40	11	20	5,841.03
16 keV	29 / 44	15	22	3.15
19.5 keV	41 / 46	5	23	5.92

Main findings

99 of 130 tests completed (76.2%); 31 were skipped for foil-movement failures. 1,105 light frames were stored: 990 from complete tests and 115 partial-test frames excluded from the online fits.

8.8 keV: reduced chi-square = 5,841. The joint attenuation model is strongly inconsistent with the measured test ratios. Its fitted thicknesses are diagnostic estimates, not reliable calibration values.

16 and 19.5 keV: reduced chi-square = 3.15 and 5.92. These blocks are much more consistent, though residual scatter still exceeds the original per-test error model. Many inferred thickness shifts repeat at both energies.

All 23 included foils remain mathematically identifiable at two or more energies despite skips. The excluded Al 204 micrometre foil, 2_6, was not calibrated. Cu 544 micrometres relies on 19.5 keV for its higher-quality estimate because it is unavailable at 16 keV.

No recorded frame exceeded the configured 1,000,000 counts/pixel/s limit. The maximum was 107,803 counts/pixel/s. This check does not independently demonstrate detector linearity.

Effective foil thicknesses

Online joint-fit results; uncertainties are one standard error (1 sigma), in micrometres.

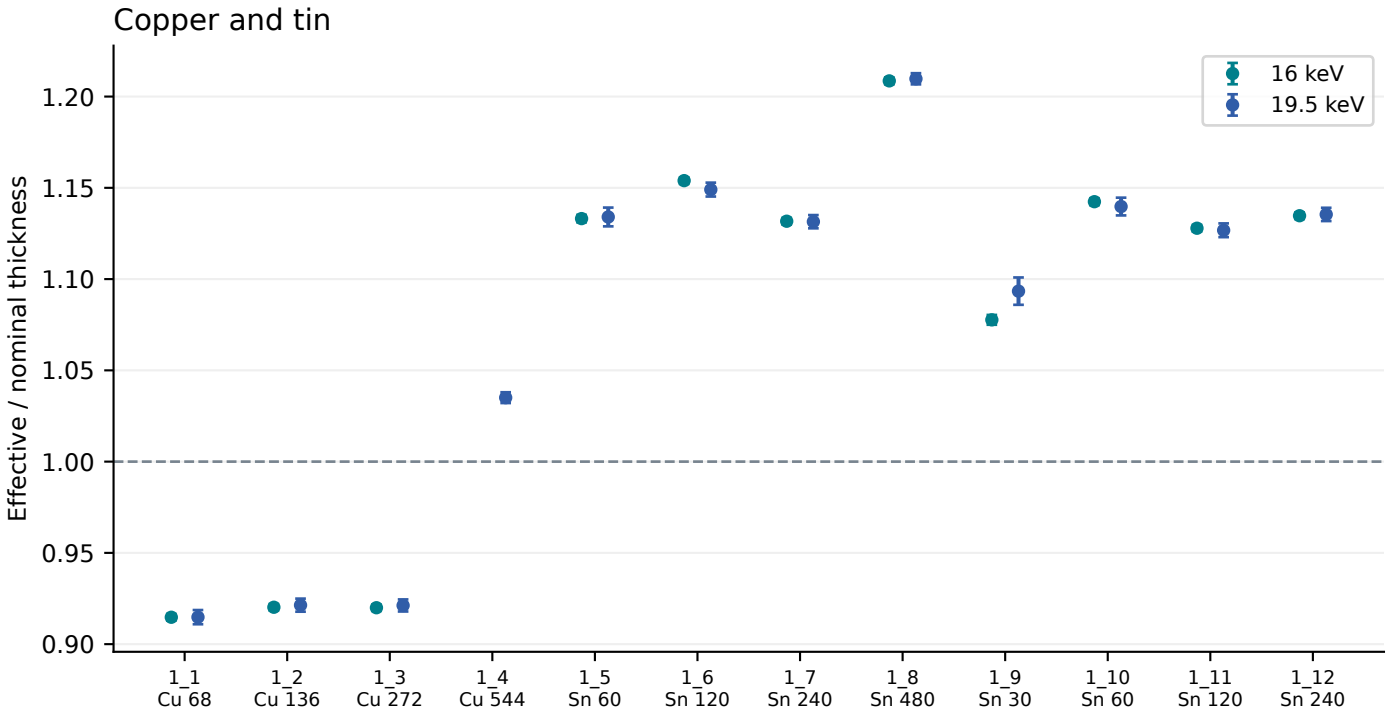
Foil	Nominal (µm)	8.8 keV*	16 keV	19.5 keV
1_1	Cu 68	68.83 ± 7.90	62.20 ± 0.12	62.21 ± 0.26
1_2	Cu 136	134.21 ± 13.75	125.15 ± 0.24	125.31 ± 0.49
1_3	Cu 272	257.31 ± 26.70	250.23 ± 0.48	250.57 ± 0.90
1_4	Cu 544	581.32 ± 44.88	—	563.07 ± 1.61
1_5	Sn 60	66.05 ± 5.95	67.99 ± 0.12	68.04 ± 0.31
1_6	Sn 120	130.23 ± 11.06	138.47 ± 0.22	137.88 ± 0.46
1_7	Sn 240	—	271.61 ± 0.45	271.55 ± 0.87
1_8	Sn 480	—	580.12 ± 0.91	580.68 ± 1.47
1_9	Sn 30	32.34 ± 2.55	32.33 ± 0.08	32.80 ± 0.22
1_10	Sn 60	66.36 ± 5.39	68.54 ± 0.11	68.39 ± 0.29
1_11	Sn 120	129.77 ± 11.16	135.34 ± 0.22	135.21 ± 0.45
1_12	Sn 240	—	272.33 ± 0.44	272.50 ± 0.87
2_1	Mo 20	20.65 ± 1.88	20.16 ± 0.06	20.00 ± 0.24
2_2	Mo 40	42.25 ± 4.23	40.67 ± 0.09	39.99 ± 0.24
2_3	Mo 80	81.05 ± 7.73	81.08 ± 0.16	79.53 ± 0.37
2_4	Mo 160	161.78 ± 13.03	182.63 ± 0.29	179.81 ± 0.64
2_5	Al 102	111.40 ± 14.03	112.21 ± 1.07	109.28 ± 3.44
2_6	Al 204	excluded	excluded	excluded
2_7	Al 408	446.25 ± 38.06	437.73 ± 1.06	443.85 ± 4.72
2_8	Al 816	865.69 ± 82.44	880.74 ± 1.52	882.64 ± 5.00
2_9	Al 9	10.78 ± 3.20	9.16 ± 1.09	14.50 ± 3.84
2_10	Al 18	21.79 ± 4.79	18.80 ± 1.17	21.81 ± 3.65
2_11	Al 36	47.33 ± 7.34	41.33 ± 0.78	40.65 ± 3.11
2_12	Al 72	67.67 ± 13.76	56.51 ± 1.30	56.19 ± 3.73

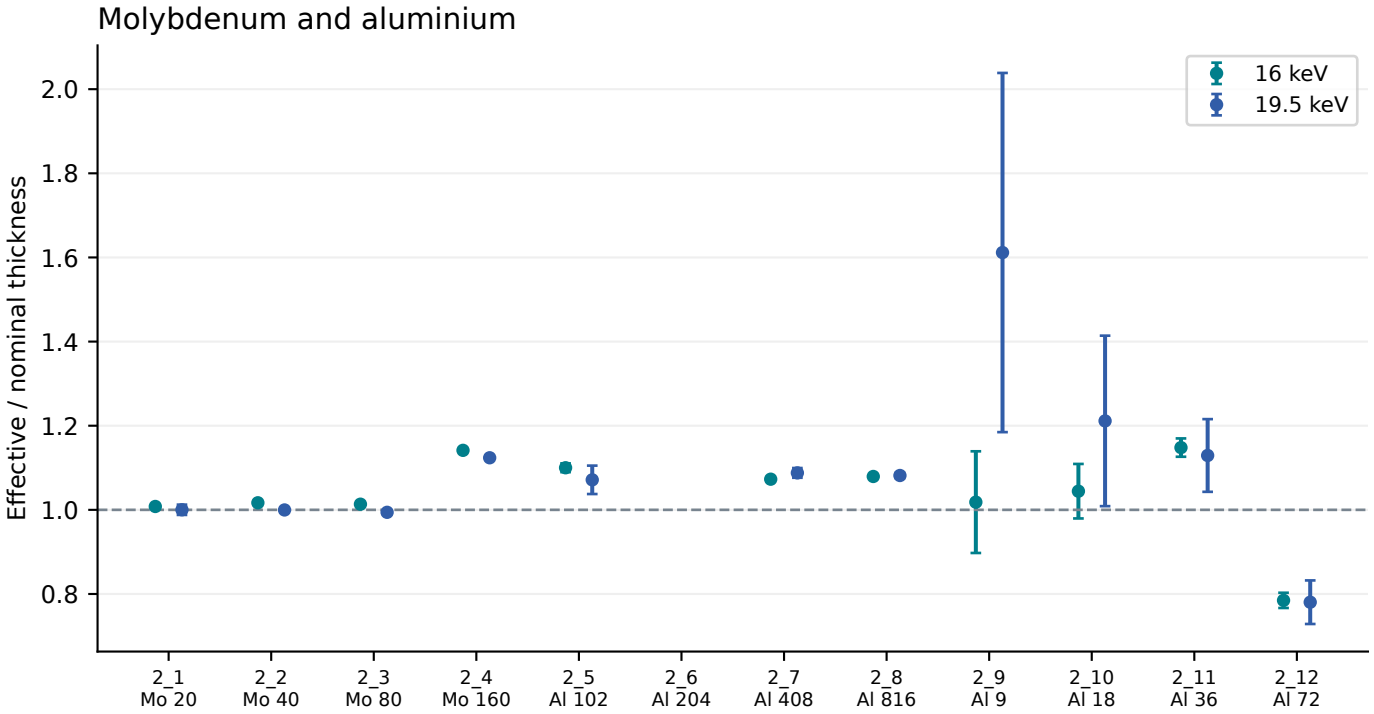
* The 8.8 keV fit has exceptionally large residuals. Its error bars were inflated by the fit's reduced chi-square; they do not cure model failure. A dash means that foil was not identifiable at that energy. Effective thickness assumes the specified material, density and monochromatic attenuation model.

Errors include the online fit covariance and residual inflation. They exclude uncertainty in the nominal absorption model and unmodeled detector, spectral, background and beam-position effects. Twenty agreeing high-energy thickness corrections were subsequently adopted; see the final calibration and follow-up pages for the current decision.

High-energy thickness patterns

Independent 16 and 19.5 keV fits; nominal thickness corresponds to scale 1.

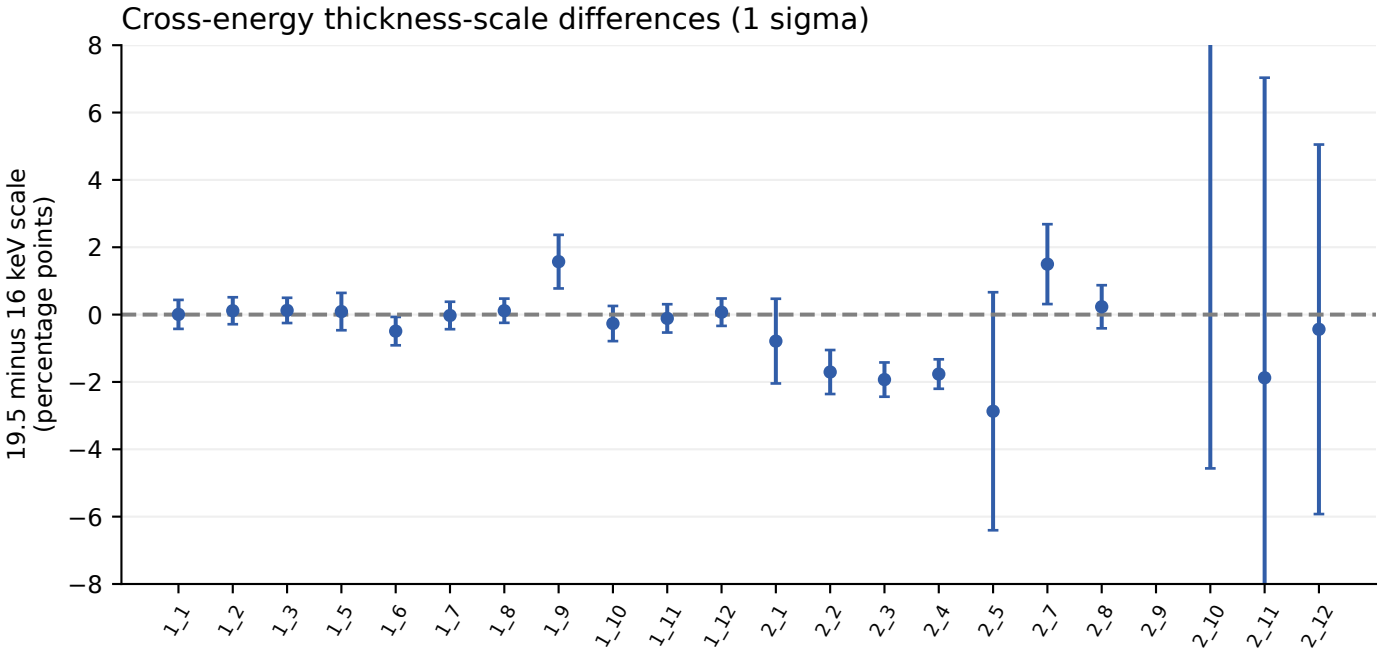




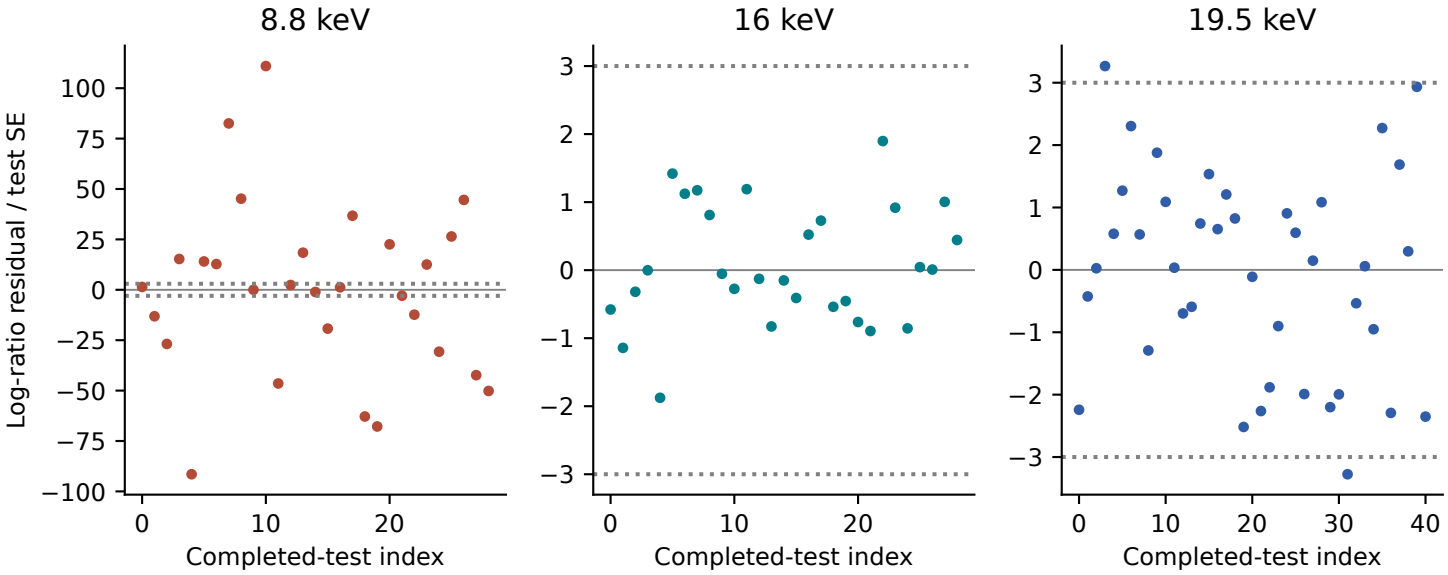
Repeatable departures: Cu 68/136/272 µm are approximately 8% thinner than nominal; most Sn foils are 8–15% thicker, with Sn 480 µm about 21% thicker. Mo 160 µm is about 12–14% thicker. Al 72 µm is near 56 µm (about 22% thinner). These are effective-thickness observations, not proof of different material or density.

Energy consistency and fit residuals

Residuals distinguish statistical precision from agreement with the attenuation model.



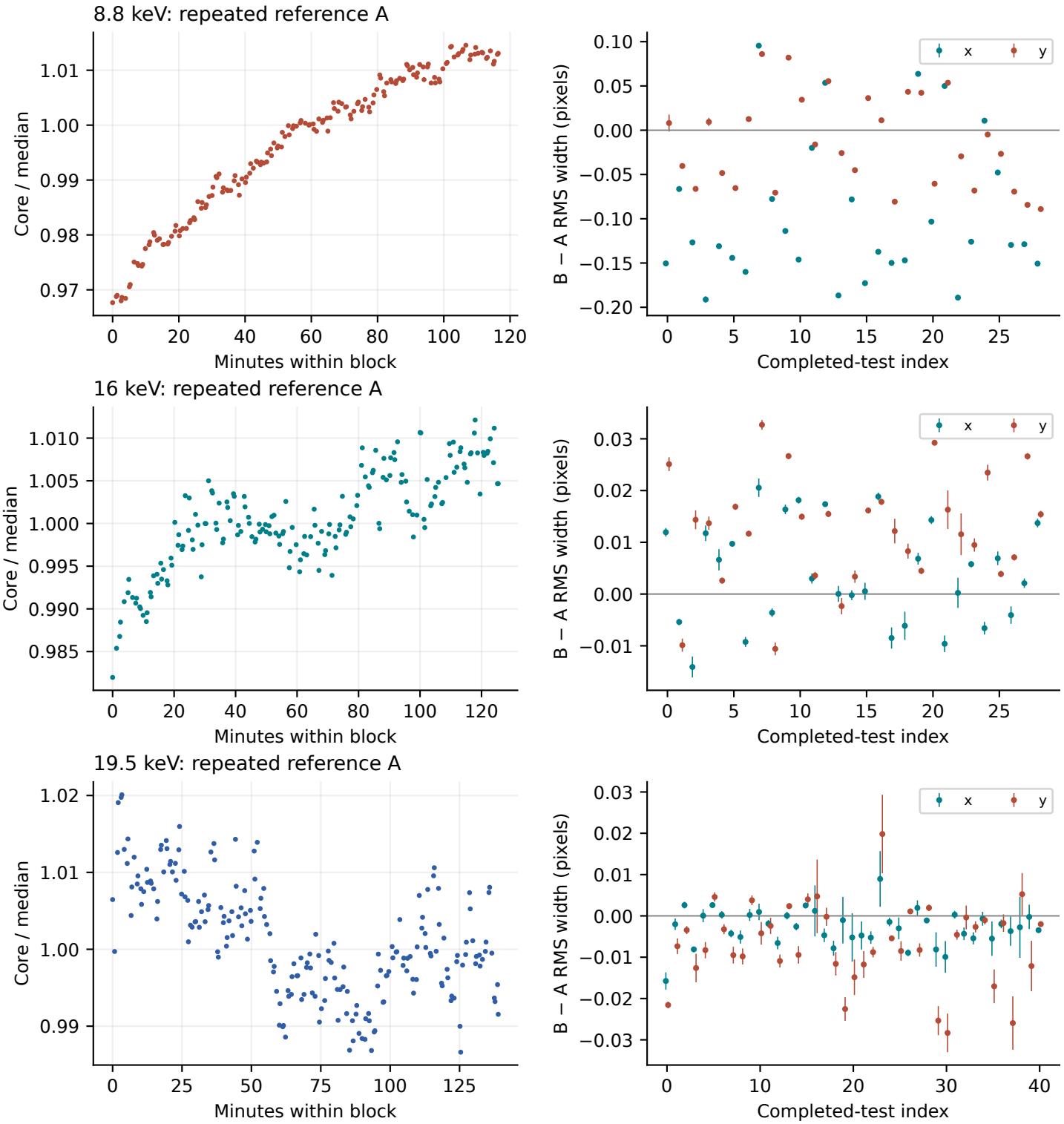
Plot limited to ±8 percentage points; the weak thin-Al differences extend beyond it. Mo 80 µm (2_3) differs by -1.93 ± 0.51 points (3.79 sigma); Mo 160 µm (2_4) by -1.77 ± 0.44 points (4.04 sigma). These individual, uncorrected comparisons warrant investigation but do not by themselves identify a composition error.



Bottom panels use each test's original ratio uncertainty, before global fit-error inflation. They show why the 8.8 keV fit should not be accepted just because its fitted optical depths are positive. The current data alone do not separate harmonics, state-dependent beam/profile effects, ROI capture, drift or detector response as the cause.

Beam profile and reference stability

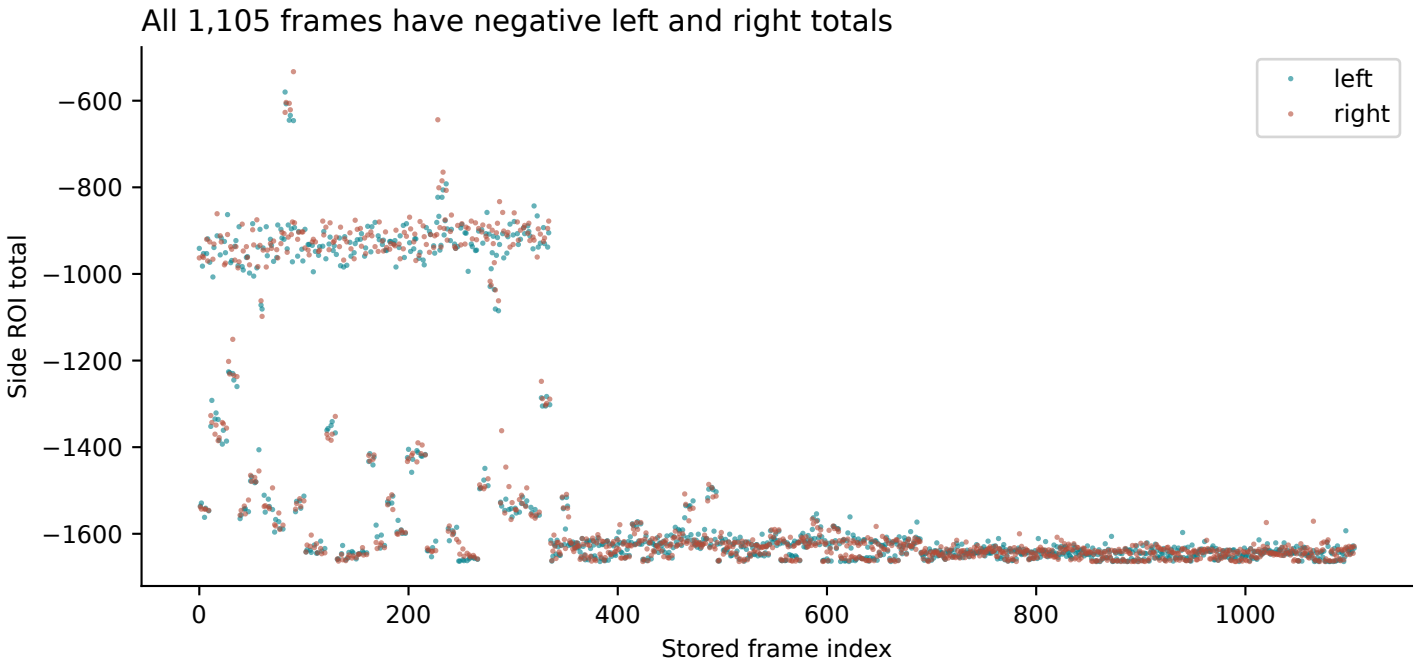
Same acquisition frames, fixed 40 × 40 pixel core ROI within each energy block.



All 99 completed-test beam-moment summaries were flagged valid. Median absolute B−A width changes (x, y) were (0.129, 0.045) pixels at 8.8 keV, (0.0069, 0.0137) at 16 keV, and (0.0028, 0.0073) at 19.5 keV. Stack substitutions change several foils, so these are stack-associated profile changes, not independent per-foil roughness measurements.

Off-axis diagnostics and acquisition quality

Interpret signed side-ROI totals before assigning diffuse-scattering meaning.



The left ROI ranged from -1,664 to -580 counts and the right from -1,664 to -533. Their medians were -1,629 and -1,630. These are not interpretable as nonnegative photon counts; masked/gap pixels or detector sentinel values are plausible contributors. The stored normalized values are diagnostic signed quantities, not a calibrated scattering signal.

No dark frames were acquired and the analysis assumed zero background. The 128 × 128 side ROIs were positioned with a 16-pixel gap from the core. Image bounds were checked, but this does not establish that ROIs avoid module gaps or bad pixels. Image-level masking and active-pixel normalization are needed for quantitative side-intensity interpretation.

Core totals ranged from 12,554 to 27,583,799 counts per 10 s frame. No frame was flagged saturated by the configured maximum-pixel-rate check. All completed tests used five pairs.

Detector settings were applied twice per energy with 30 s waits. Energy and threshold readbacks matched 8.8/4.4, 16/8.0 and 19.5/9.75 keV. Gain-menu setting was 1 throughout; the independent gain readback was unavailable, and that fact was recorded. This report reads scalar streams and metadata; it does not inspect the TIFF image arrays.

Movement failures and follow-up

Recovery kept acquisition running; partial-test frames remain available for targeted reanalysis.

Foil named in failure	Number of skipped-test records
1_9	9
1_8	6
2_9	5
2_7	2
2_2	2
1_2	2
1_5	2
2_12	1
1_10	1
1_7	1
1_6	1

31 tests were skipped, all classified as foil movement failures. The counts above total 32 foil mentions because one test reported two failed foils. They are not failure probabilities: the reference foils were commanded more often than others. Some final snapshots reached the requested position after a status timeout; those tests were still excluded as designed.

1. Investigate 8.8 keV before adopting its calibration.

Check the largest residual tests against the repeated-reference intensity and profile history, plus images/ROI capture and spectral contamination. Repetition alone will not fix a systematic model discrepancy.

2. Use high-energy thicknesses as candidate results.

The repeated Cu/Sn/Al shifts are useful evidence. Verify held-out combinations and the Mo 80/160 µm energy differences before updating production tables.

3. Address movement reliability.

Prioritize 1_9, 1_8 and 2_9; keep 2_6 excluded until repaired. Review saved failure details before deciding whether any late-settled partial pairs can be recovered offline.

4. Rework side-intensity reduction.

Inspect TIFF frames for masked/gap pixels and integrate over valid pixels. The negative totals should not be used to claim foil scattering or roughness.

Skipped-test audit trail

All 31 skipped tests; pair is displayed as 1-based. Full messages/readbacks are stored in Tiled.

Test ID	Failed pair/state	Complete pairs before skip	Foil(s) named
E8800_001	1 / A	0	2_9
E8800_002	4 / A	3	2_9
E8800_005	1 / B	0	2_7
E8800_008	2 / A	1	2_2
E8800_010	1 / A	0	2_12
E8800_020	4 / A	3	2_9
E8800_022	1 / B	0	2_7
E8800_023	5 / B	4	2_9
E8800_025	5 / B	4	2_9
E8800_026	1 / A	0	2_2
E8800_037	5 / B	4	1_10
E16000_000	2 / A	1	1_9
E16000_001	4 / A	3	1_9
E16000_002	4 / A	3	1_2
E16000_005	5 / B	4	1_8
E16000_009	1 / B	0	1_9
E16000_017	1 / B	0	1_5
E16000_018	1 / B	0	1_8
E16000_020	3 / B	2	1_8
E16000_021	2 / A	1	1_9
E16000_022	5 / B	4	1_5
E16000_023	2 / A	1	1_9
E16000_033	4 / A	3	1_9
E16000_036	4 / A	3	1_9
E16000_040	1 / B	0	1_9
E16000_041	1 / A	0	1_9
E19500_000	1 / B	0	1_7, 1_8
E19500_009	1 / B	0	1_8
E19500_015	1 / A	0	1_2
E19500_029	2 / A	1	1_6
E19500_034	1 / A	0	1_8

Source: NSLS-II Tiled, smi/raw, run 7cd58d5e-7f31-4fe4-b64f-aad0bde951db. Streams: primary, ab_summary, foil_summary, fit_diagnostics, skipped_tests, shape_summary, detector_settings and roi_settings. Report generated 2026-10-19 12:49 UTC.

Follow-up: scan 1171325

UID 4348a125-4a82-414e-9c5c-15c1513303e7 | Isolated reference-plus-target comparisons

The follow-up completed in 13.06 minutes with 5 of 6 tests completed and 21 light frames stored. Twenty frames belong to complete tests; one reference frame belongs to the skipped 19.5 keV test of 2_6. Each completed test used two A/B pairs and 10 s frames.

Foil	16 keV (µm)	19.5 keV (µm)	Agreement
2_3 / Mo 80	80.86487 ± 0.06929	80.08146 ± 0.19861	3.72 sigma: fails
2_4 / Mo 160	181.94951 ± 0.20832	180.19298 ± 0.34920	4.32 sigma: fails
2_6 / Al 204	219.11592 ± 0.55038	Insertion failed	One energy only

The same 3-combined-standard-error rule was applied to the two-energy comparison. Neither Mo foil passes it. The new Al result establishes a baseline about 7.41% above nominal, but its missing second-energy result prevents a two-energy correction.

The Mo differences have the same direction as in scan 1171324. Their magnitudes are smaller but remain statistically significant under the recorded error model. A fixed reference cancels in each A/B ratio, so this result does not depend on assuming corrected reference thicknesses are exact. It does not identify the physical source of the energy dependence.

At 16 keV there are three equations for three target optical depths; at 19.5 keV, two for two. Both fits have zero residual degrees of freedom. Their nearly zero chi-square is therefore expected and must not be read as validation. Two pairs provide only limited evidence about empirical repeatability and systematic errors.

All five beam-moment summaries were valid; no saturation flags were recorded. Maximum pixel rate was 9,704 counts/pixel/s. Side totals remained negative: left -1662 to -1592, right -1663 to -1591. The only skipped test was followup_E19500_2_6: f6 failed insertion on the first B frame. The recorded state confirmed 2_6 was still retracted.

Active thickness calibration

Version smi-attenuators-1171324-v1 • 20 accepted foils • unchanged after follow-up

Foil	Nominal (µm)	Applied (µm)	Sigma (µm)	Source
1_1	Cu 68	62.2028	0.1128	measured
1_2	Cu 136	125.1809	0.2183	measured
1_3	Cu 272	250.3015	0.4224	measured
1_4	Cu 544	544.0000	—	nominal fallback
1_5	Sn 60	67.9958	0.1152	measured
1_6	Sn 120	138.3641	0.2287	measured
1_7	Sn 240	271.5967	0.3965	measured
1_8	Sn 480	580.2771	0.7726	measured
1_9	Sn 30	32.3842	0.1501	measured
1_10	Sn 60	68.5231	0.1063	measured
1_11	Sn 120	135.3177	0.2007	measured
1_12	Sn 240	272.3666	0.3921	measured
2_1	Mo 20	20.1513	0.0590	measured
2_2	Mo 40	40.5880	0.2238	measured
2_3	Mo 80	80.0000	—	nominal fallback
2_4	Mo 160	160.0000	—	nominal fallback
2_5	Al 102	111.9513	1.0178	measured
2_6	Al 204	204.0000	—	nominal fallback
2_7	Al 408	438.0288	1.3115	measured
2_8	Al 816	880.9030	1.4547	measured
2_9	Al 9	9.5611	1.3996	measured
2_10	Al 18	19.0768	1.1107	measured
2_11	Al 36	41.2877	0.7606	measured
2_12	Al 72	56.4725	1.2300	measured

Accepted results combine the original 16 and 19.5 keV marginal thickness estimates using inverse-variance weights. Their 1-sigma errors include the original energy-fit inflation and additional two-point scatter inflation where needed. Cross-foil errors remain correlated; the table is not an independent-error model for stack transmission.

Applied optical depth = nominal optical depth × effective/nominal thickness. Base material tables remain intact. Other energies use thickness-model extrapolation. No follow-up target qualifies for an additional correction under the same acceptance rule.

Remaining issues and next measurements

Combined assessment of scans 1171324 and 1171325

2_6: baseline obtained; insertion remains unreliable.

The 16 keV measurement is 219.11592 ± 0.55038 µm. Check/repair insertion, then repeat only its 19.5 keV reference-plus-foil test. If intervention changes the foil or illuminated position, repeat 16 keV as well. It remains excluded from normal commissioning designs.

2_3 and 2_4: persistent energy dependence.

The isolated follow-up confirms disagreement at 3.72 and 4.32 sigma. Repeat with a different reference stack and a controlled threshold comparison at fixed energy, retaining raw images. Investigate spectral contamination, detector response, background and positioning before assigning a different composition. Both retain nominal thickness in the active model.

1_4: still lacks a usable independent energy check.

The original 19.5 keV estimate remains 563.07 ± 1.61 µm; nominal 544 µm remains applied. Use the documented 19/19.5 keV substitution comparison and propagate reference covariance. The close energy separation is a weak material test.

Thin Al and mechanical reliability.

Accepted 2_9 and 2_10 thicknesses have approximately 15% and 6% uncertainty. The original intermittent failures of 1_9, 1_8 and 2_9 were not re-tested here. Their existing thickness estimates do not establish motion reliability.

8.8 keV and side-ROI interpretation remain open.

No follow-up data address the original extreme 8.8 keV residuals. Both runs have negative side totals requiring TIFF-level masking/background review before quantitative scattering use.

Data provenance: NSLS-II Tiled smi/raw, both full UIDs recorded in this report. Follow-up baselines are recorded separately in attenuator_calibration.py. See docs/commissioning/CALIBRATION_1171324.md for adoption criteria and run commands.